

ON THE DISTRIBUTION AND POWER OF A TEST FOR A SINGLE OUTLIER

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Synopsis. The distributions under null and alternative hypotheses of a well-known single outlier test are derived.

1. Notation and Introduction

Suppose that $y_1, y_2, \dots, y_n$ are a random sample from a $N(\xi, \sigma^2)$ population and that u is a $\sigma^2 \chi^2_\nu$ variate independent of the set of y_i .

Let $y_{(1)} < y_{(2)} < \dots < y_{(n)}$ denote the y values ranked in ascending order. Define

$$\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$$

$$s^2 = \sum_{i=1}^n (y_i - \bar{y})^2$$

$$S^2 = s^2 + u$$

$$\bar{y}_j = \frac{1}{n-1} \sum_{\substack{i=1 \\ i \neq j}}^n y_i$$

$$s_j^2 = s^2 + u$$

$$\bar{y}_{(j)} = \frac{1}{n-1} \sum_{\substack{i=1 \\ i \neq j}}^n y_{(i)}$$

$$s_{(j)}^2 = \sum_{\substack{i=1 \\ i \neq j}}^n (y_{(i)} - \bar{y}_{(j)})^2$$

$$S_{(j)}^2 = s_{(j)}^2 + u$$

Various procedures have been proposed for testing $y_{(n)}$ as an outlier. One is based on the test statistic

$$B = \frac{y_{(n)} - \bar{y}}{S}$$

This is a generalisation by Quesenberry and David (1961) of a test statistic originally proposed by Pearson and Chandrasekhar (1936). It has been shown by Paulson (1952) to have certain properties of optimality.

Another intuitively reasonable test statistic is

$$R = \frac{y(n) - \bar{y}(n)}{S(n)}$$

This is a generalisation of a test statistic by Doornbos (1959). It is shown by Quesenberry and David that R and B are mathematically equivalent, in fact:

$$B = \frac{(n-1)R}{n \sqrt{1 + (n-1)R^2/n}}$$

The range of R is easily seen to be $(0, \infty)$. The range of B is bounded above. Using the equation connecting R and B we see that the range of B is $(0, \sqrt{(n-1)/n})$.

Quesenberry and David derived approximations to the high fractiles of B . In this paper the exact distribution of R (and hence B) will be derived.

2. Derivation

Since y_i and $\bar{y}_i$ are independent,

$$y_i - \bar{y} \sim N \left[0, \left(1 + \frac{1}{n-1}\right) \sigma^2 \right]$$

or
$$\sqrt{\frac{n-1}{n}} \frac{y_i - \bar{y}_i}{\sigma} \sim N[0, 1].$$

Also s_i^2 and u are independent of both $\bar{y}_i$ and y_i .

By standard theory,

$$S_i^2 = s_i^2 + u \sim \sigma^2 \chi_{n+\nu-2}^2$$

Thus, letting

$$c = \sqrt{\frac{n}{(n-1)(n+\nu-2)}}$$

$$\frac{1}{c} \frac{y_i - \bar{y}_i}{S_i} \sim t_{n+\nu-2}$$

Thus, defining

$$r_i = \frac{y_i - \bar{y}_i}{S_i}$$

we have that

$$(1) \quad r_i \sim c t_{n+\nu-2}$$

Now consider the set $y_1, y_2, \dots, y_{n-1}$.

This subsample has mean $\bar{y}_n$ and sum of squared deviations s_n^2 . Apply the transformation

$$y_i = \bar{y}_n + b_i \cdot S_n, \quad i = 1 \text{ to } n-1$$

This is a slight modification of a standard transformation traced back to Behrens (1929).

Quesenberry and David (1961) proved that an arbitrary $b_i, i = 1, \dots, n-1$, is distributed independently of $(y_n - \bar{y})/S$. A minor adaptation of this proof shows that the set b_i is independent of $(y_n - \bar{y})/S$. But, as shown in Quesenberry and David's article, the latter quantity is simply

$$\frac{(n-1)r_n}{n\sqrt{1 + (n-1)r_n^2/n}}$$

where $r_n = (y_n - \bar{y}_n)/S_n$.

Hence

(2) r_n is independent of the set $b_i, i = 1, \dots, n-1$.

Let r and dr be preassigned.

Now consider the conditional probability

(3) $P[r < r_n < r + dr \mid r_n < \max b_i]$

This is of the form $P[A|B]$, and so may be written as

$$\begin{aligned} & \frac{P[A \text{ and } B]}{P[B]} \\ &= \frac{P[r < r_n < r + dr \text{ and } (\max b_i < r_n)]}{P[\max b_i < r_n]} \end{aligned}$$

If dr is small then it is clear that

$$\begin{aligned} P[B|A] &= P[\max b_i < r_n \mid A] \\ &= P[\max b_i < r \mid A] + O(dr) \end{aligned}$$

Thus the numerator term is:

$$P[r < r_n < r + dr \text{ and } (\max b_i < r)] + o(dr)$$

so equation 3 becomes

$$(4) \quad \frac{P[r < r_n < r + dr \text{ and } (\max b_i < r)] + o(dr)}{P[\max b_i < r_n]}$$

In the numerator term we note the following:

- (a) r_n is independent of the set b_i
- (b) the b_i define a set of statistics of the form $(y_i - \bar{y})/S$ from a sample of size $n-1$. Thus $\max b_i$ is a B statistic from a sample of size $n-1$.

Let $G_{n-1, \nu}(x)$ denote the cumulative distribution function of a B statistic from a sample of size n with ν degrees of freedom for u . Then it follows that the numerator term of (4) may be written

$$P[r < r_n < r + dr] P[\max b_i < r] + o(dr)$$

$$(5) \quad = f(r)dr \cdot G_{n-1, \nu}(r) + o(dr)$$

where $f(r)$ denotes the density of the distribution of r_n which is given by (1) above.

The denominator term in (4) is simply the probability that y_n is the largest of the set of y values. By elementary combinatorial theory, this is $1/n$.

Thus the conditional probability (3) becomes

$$(6) \quad nf(r)G_{n-1,\nu}(r)dr + o(dr)$$

Now the y_i are independently and identically distributed. Further, if $\max b_i < r_n$ then $r_n = R$

Thus the conditional probability (3) is

$$(7). \quad P[r < R < r + dr]$$

Thus from (6) the density of R is

$$\lim_{dr \rightarrow 0} \frac{nf(r)G_{n-1,\nu}(r)dr + o(dr)}{dr}$$

$$(8) \quad = nf(r)G_{n-1,\nu}(r)$$

Let $F_{n,\nu}(x)$ denote the cumulative distribution function of R . Then by the equivalence of R and B we have

$$G_{n,\nu}(x) = F_{n,\nu} \left[\frac{nx/(n-1)}{\sqrt{1-nx^2/(n-1)}} \right], x < \sqrt{\frac{n-1}{n}}$$

$$(9) \quad = 1, \quad x \geq \sqrt{\frac{n-1}{n}}$$

Thus, finally

$$\begin{aligned} F_{n,\nu}(x) &= \int_0^x nf(r) G_{n-1,\nu}(r) dr \\ &= \int_0^x nf(r) F_{n-1,\nu} \left[\frac{nx/(n-1)}{\sqrt{1-(n-1)r^2/(n-2)}} \right] dr \end{aligned}$$

where it is understood that where $r > \sqrt{(n-2)/(n-1)}$ then the $F_{n-1,\nu}$ term is taken to be 1.

This is the fundamental result defining the distribution of R . It defines a straightforward recursive technique for finding $F_{n,\nu}(x)$ in terms of elementary functions or computing its numerical value.

A starting value is required: let $n = 2$, then

$$r_2 = \frac{y_2 - y_1}{\sqrt{O+u}}$$

and

$$y_2 - y_1 \sim N(0, 2\sigma^2)$$

and independently

$$u \sim \sigma^2 \chi^2$$

Thus

$$\sqrt{\frac{\nu}{2}} r_2 \sim t_\nu$$

and

$$R = |r_2|$$

(11) $F_{2, \nu}(x)$ is simply the probability that a t_ν variate exceeds $x\sqrt{\nu/2}$ in modulus.

3. The performance of the test

Now let us suppose that a single contaminant is present. For convenience we shall assume that y_n , the last observation drawn, is a member of a different population from its temporal predecessors. Our alternative hypothesis is:

$$H_1 : t_1, t_2, \dots, y_{n-1} \sim N(\xi, \sigma^2) \\ y_n \sim N(\xi + \delta\sigma, \lambda^2 \sigma^2)$$

so that the contaminant y_n has a location shift and a scale change.

Several criteria have been proposed for assessing the performance of the test in detecting contamination and identifying the contaminant.

One is the power function: the probability of rejecting H_0 given H_1 . This is generally unsatisfactory for our purpose, since mere rejection of H_0 does not guarantee the correct identification of the contaminant.

Let k satisfy

$$P[R \geq k | H_0] = \alpha$$

Then three other measures of performance which may be useful are:

- (1) $\beta_1 = P[r_n \geq k]$
- (2) $\beta_2 = P[r_n \geq k \text{ and } \max b_i < r_n]$
- (3) $\beta_3 = P[r_n \geq k | \max b_i < r_n]$

β_2 is of particular interest — it is the probability that a contaminant, if present, is correctly identified as such. As we will show below, β_3 is simply obtained from β_2 and is not essentially different. β_1 has no useful practical interpretation — it has been used as an approximation to β_2 by David and Paulson (1965). In the case $\lambda = 1$ (i.e. the contaminant has a location shift only), David and Paulson have derived an exact expression for β_1 . Exact expressions for $\beta_1, \beta_2, \beta_3$ in the general case $\lambda \geq 1$ will be derived below.

As before we note that

$$\bar{y}_n \sim N[\xi, \sigma^2/(n-1)] \text{ and}$$

$$S_n^2 \sim \sigma^2 \chi_{n+\nu-2}^2 \text{ and independently}$$